Radiometry and Detection Signal Flow in VibeVolts

Simulation Reference Documentation

June 28, 2026

1 Introduction

This document outlines the step-by-step mathematical transformations used in the VibeVolts codebase to calculate the target signal, background noise, and resulting Signal-to-Noise Ratio (SNR) on a satellite or observatory detector.

2 Step 1: Solar Flux Calculation

The source of illumination is the Sun. The function `fluxes(band)` in `radiometry_calcs.py` retrieves the solar apparent magnitude m_{sun} and the zero-point calibration flux ZP (flux of a 0-magnitude object in photons/s/m^2) for a given band.

The solar flux at Earth/target F_{sun} is calculated as:

$$F_{\text{sun}} = 10^{-0.4 \cdot m_{\text{sun}}} \cdot \text{ZP} \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{m}^2}\right] \tag{1}$$

3 Step 2: Target Reflection (Lambertian Sphere)

The target is modeled as a diffuse Lambertian sphere of radius R and albedo a. The phase angle α is the physical angle between the Sun (light source) and the observer (satellite detector) as seen from the target's center.

The phase function $\Phi(\alpha)$ for a Lambertian sphere is:

$$\Phi(\alpha) = \frac{2}{3\pi^2}\left[\sin\alpha + (\pi - \alpha)\cos\alpha\right] \tag{2}$$

The intensity $I(\alpha)$ of the scattered light in the direction of the observer is:

$$I(\alpha) = F_{\text{sun}} \cdot a \cdot (\pi R^2) \cdot \pi\Phi(\alpha) \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{sr}}\right] \tag{3}$$

Inside `lambertian.py`'s `lambertiansphere` function, the returned value (often called `target_brightness` or $J(\alpha)$) is:

$$J(\alpha) = F_{\text{sun}} \cdot \underbrace{\left(a \cdot (\pi R^2) \cdot \frac{2}{3\pi}\left[\sin\alpha + (\pi - \alpha)\cos\alpha\right]\right)}_{\text{effective\_cross\_section}} = \pi \cdot I(\alpha) \tag{4}$$

4 Step 3: Apparent Flux at the Detector

As the reflected light propagates over distance d to the observer, the flux spreads over a sphere's surface. The apparent photon flux (irradiance) at the entrance aperture of the detector, F_{det}, is computed in `scandetectors.py` as:

$$F_{\text{det}} = \frac{J(\alpha)}{\pi d^2} = \frac{I(\alpha)}{d^2} = \frac{2aF_{\text{sun}}R^2}{3\pi d^2}\left[\sin\alpha + (\pi - \alpha)\cos\alpha\right] \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{m}^2}\right] \tag{5}$$

5 Step 4: Target Signal Collection

The total number of target signal photons S collected by the sensor during the integration time t is calculated in `scandetectors.py` as:

$$S = F_{\text{det}} \cdot t \cdot A \cdot \eta \cdot f \quad \text{[photons]} \tag{6}$$

Where:

- t: Integration (exposure) time in seconds (`integrationTime`).
- A: Detector aperture entrance area in m^2 (`apertureArea`).
- η: Overall system quantum efficiency (`qe`).
- f: Photometric efficiency fraction inside the photometry bucket (`photoEff`).

6 Step 5: Background Noise Collection

The background sky/space brightness F_{bg} is given in photons/s/sr/m^2 (derived from the filter band's zero point and magnitudes per square arcsecond).

Assuming background-dominated noise (shot noise), the background photons B collected by a pixel of solid angle Ω (in steradians) is:

$$B = F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta \quad \text{[photons]} \tag{7}$$

The background shot noise N is the standard deviation of this Poisson process:

$$N = \sqrt{B} = \sqrt{F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta} \quad \text{[photons]} \tag{8}$$

7 Step 6: Signal-to-Noise Ratio (SNR) and Integration Time

The SNR γ is:

$$\gamma = \frac{S}{N} = \frac{F_{\text{det}} \cdot t \cdot A \cdot \eta \cdot f}{\sqrt{F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta}} = F_{\text{det}} \cdot f \cdot \sqrt{\frac{t \cdot A \cdot \eta}{F_{\text{bg}} \cdot \Omega}} \tag{9}$$

Solving this expression for the required integration time t to achieve a target SNR γ yields:

$$t = \gamma^2 \cdot \frac{F_{\text{bg}} \cdot \Omega}{F_{\text{det}}^2 \cdot A \cdot \eta \cdot f^2} \quad \text{[seconds]} \tag{10}$$

This is the corrected equation implemented in `detector.py`'s `requiredIntegrationTime`.

8 Summary of Variable Definitions

Variable	Code Representation	Description
F_{sun}	`sun`	Incident solar flux (photons/s/m^2)
R	`radius`	Radius of the sphere target (m)
a	`albedo`	Reflected fraction of target surface ($0.0 \ldots 1.0$)
α	`angle_light_observer`	Phase angle (radians)
d	`hit_norms`	Target-to-detector distance (m)
F_{det}	`detectorFlux`	Incident target flux on detector (photons/s/m^2)
A	`apertureArea`	Entrance aperture area (m^2)
t	`integrationTime`	Exposure time (seconds)
η	`qe`	System quantum efficiency ($0.0 \ldots 1.0$)
f	`photoEff` / `photfrac`	Photometric bucket fraction ($0.0 \ldots 1.0$)
F_{bg}	`space` / `sky`	Background sky radiance (photons/s/sr/m^2)
Ω	`pixelOmega`	Pixel solid angle (steradians)

Table 1: Variable mappings between the mathematical equations and the python code.